

Completing the Square

One structure. Fill the blanks. Read the turning point straight off it.



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THE TOPIC ON ONE MAP

Completing the Square

01 – The two-step method

$$(x + p)^2 + q$$

$$a(x + b)^2 + c$$

02 – Negative a

$$28 + 24x - 6x^2$$

$$a - b(x - c)^2$$

03 – Read the turning point

minimum / maximum

line of symmetry

sketch

04 – The “hence” questions

hence solve

surd form

transformations

Four ideas. After each idea you practise it straight away: my questions first, then selected practice, then real past papers.

01 The Two-Step Method

Any quadratic $ax^2 + bx + c$ — list four numbers, then drop them into one structure.

WHEN YOU SEE → YOU DO

“in the form $(x + p)^2 + q$ ”

→ complete the square

“express ... in the form $a(x + b)^2 + c$ ”

→ list $a, b, c, \frac{b}{2a}$ → structure

“where a, b and c are integers”

→ match the letters to your answer, not to the question

1

List your elements

$$a = ?$$

$$b = ?$$

$$c = ?$$

$$\frac{b}{2a} = ?$$

Always write $ax^2 + bx + c$ in this order first, even when the question writes it backwards.

2

Use the structure

Write the same structure every time — the **same** $\frac{b}{2a}$ goes in **both** brackets.

$$\underbrace{a\left(x + \frac{b}{2a}\right)^2}_{\text{keep it as it is}} \quad - \underbrace{a\left(\frac{b}{2a}\right)^2 + c}_{\text{one number: use the calculator}}$$

WORKED MODEL • $a = 1$

Write $x^2 - 6x + 11$ in the form $(x - p)^2 + q$.

1 $a = 1 \quad b = -6 \quad c = 11 \quad \frac{b}{2a} = \frac{-6}{2} = -3$

list

2 $(x - 3)^2 - 1(-3)^2 + 11$

structure

3 $(x - 3)^2 - 9 + 11$

calculator: right part only

ANSWER $(x - 3)^2 + 2$

WORKED MODEL • $a \neq 1$

Write $2x^2 + 8x + 5$ in the form $a(x + b)^2 + c$.

1 $a = 2 \quad b = 8 \quad c = 5 \quad \frac{b}{2a} = \frac{8}{4} = 2$

list

2 $2(x + 2)^2 - 2(2)^2 + 5$

structure

3 $2(x + 2)^2 - 8 + 5$

calculator: right part only

ANSWER $2(x + 2)^2 - 3$

× **TRAP** Writing $-(2)^2$ instead of $-a(2)^2$. The a multiplies the second bracket too.

× **TRAP** The question's letters are not your a, b, c . In " $a(x + b)^2 + c$ ", the answer's b is $\frac{b}{2a}$ — read the form, then match.

IDEA 01

Practise: The Two-Step Method

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Step 1 first · $a = 1$

CH 2 · TOPIC 5 2 marks

Write $x^2 + 6x + 11$ in the form $(x + p)^2 + q$.

List first: $a =$ $b =$ $c =$ $\frac{b}{2a} =$

Answer _____

Q2 DR ESLAM $a \neq 1$ · the trap

CH 2 · TOPIC 5 3 marks

Write $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$.

The second bracket is multiplied by a too.

Answer _____

SELECTED PRACTICE

Q3

PRACTICE

Practice · Medium

CH 2 · TOPIC 5 2 marks

1 Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$ where a and b are integers.

(2 marks)

Q4

PRACTICE

Practice · Medium

CH 2 · TOPIC 5 3 marks

6 Write $x^2 - 6x + 11$ in the form $(x - a)^2 + b$

(3 marks)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 5 3 marks

10

$$x^2 + 4x - 9 = (x + a)^2 + b$$

Find the value of a and the value of b . $a = \dots\dots\dots$ $b = \dots\dots\dots$

(3 marks)

Q6**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 5 3 marks

7 $2x^2 - 6x + 5$ can be written in the form $a(x - b)^2 + c$ where a , b and c are positive numbers. Work out the values of a , b and c . $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$

(3 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q19

CH 2 · TOPIC 5 3 marks

19 Given that $\left(\sqrt[3]{\frac{1}{x}}\right)^4 = x^m$

(a) find the value of m

$m = \dots\dots\dots$

Given that a , b and c are integers,

(b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

$\dots\dots\dots$

(Total for Question 19 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q19

CH 2 · TOPIC 5 3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

$\dots\dots\dots$

(Total for Question 19 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q21

CH 2 · TOPIC 5 3 marks

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

.....

(Total for Question 21 is 3 marks)

02 Negative a ? Same structure.

No new method. Rewrite in the order $ax^2 + bx + c$ and carry the sign of a everywhere.

WHEN YOU SEE → YOU DO

“in the form $a - b(x - c)^2$ ”

→ negative a : same structure, write the number first

“ $28 + 24x - 6x^2$ ”

→ reorder to $-6x^2 + 24x + 28$ first

1

Reorder

$28 + 24x - 6x^2 \rightarrow -6x^2 + 24x + 28$, so $a = -6$, $b = 24$, $c = 28$.

2

Same structure, sign of a included

$-a(\dots)^2$ with $a = -6$ becomes $+6(\dots)^2$ — the minus-minus turns into plus.

EXAM FORM $a - b(x - c)^2$ • JUNE 2025 P2H

1 $\frac{b}{2a} = \frac{24}{-12} = -2$

list

2 $-6(x - 2)^2 - (-6)(-2)^2 + 28$

structure

3 $-6(x - 2)^2 + 24 + 28$

calculator

ANSWER $52 - 6(x - 2)^2$ so $a = 52$, $b = 6$, $c = 2$

× **TRAP** Answering in the wrong order. The form asks for “number first”: write $52 - 6(x - 2)^2$, not $-6(x - 2)^2 + 52$.

IDEA 02

Practise: Negative a

My questions first, then selected practice, then real past papers.

Q10 DR ESLAM Negative $a = -1$

CH 2 · TOPIC 5 3 marks

Write $10 - 6x - x^2$ in the form $p - (x + q)^2$.

Reorder first: $-x^2 - 6x + 10$, so $a = -1$.

Answer _____

Q11 DR ESLAM Negative $a \neq -1$

CH 2 · TOPIC 5 3 marks

Write $5 + 8x - 2x^2$ in the form $a - b(x - c)^2$.

Answer _____

SELECTED PRACTICE

Q12 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 5 2 marks

4 The function f is such that $f(x) = 5 + 6x - x^2$ for $x \leq 3$

Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

(2 marks)

Q13 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 5 4 marks

5 Write $5 + 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(4 marks)

Q14**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 5 2 marks

6 Express $7 - 4x - x^2$ in the form $p - (x + q)^2$ where p and q are constants.

(2 marks)

PAST PAPERS**Q15****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q23

CH 2 · TOPIC 5 3 marks

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(Total for Question 23 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q20

CH 2 · TOPIC 5 4 marks

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

C is the curve with equation $y = 7 + 12x - 3x^2$

The point A is the maximum point on C

(b) Use your answer to part (a) to write down the coordinates of A

(.....,)

(Total for Question 20 is 4 marks)

Q17 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1H · Q24

CH 2 · TOPIC 5 4 marks

24 Express each of a , b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

Q18 PAST PAPER Edexcel 4MA1 · May 2024 · Paper 1H · Q25

CH 2 · TOPIC 5 4 marks

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

$$f(x) = \dots\dots\dots$$

(Total for Question 25 is 4 marks)

03 Read the Turning Point

From $a(x + p)^2 + q$ the turning point is $(-p, q)$ — **flip** the inside, **keep** the outside.

WHEN YOU SEE → YOU DO

“turning point”, “minimum point”, “vertex”

→ flip the inside, keep the outside

“maximum point”

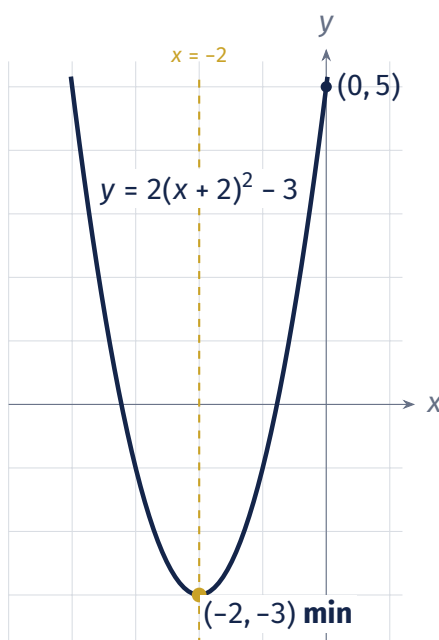
→ $a < 0$ — cap shape

“line of symmetry”

→ x = the flipped inside number

“sketch ... show the coordinates”

→ turning point + y-cut + shape



Inside $+2 \rightarrow \text{flip} \rightarrow x = -2$

Outside $-3 \rightarrow \text{keep} \rightarrow y = -3$

Shape $a = 2 > 0 \rightarrow \text{minimum}$

Symmetry the line $x = -2$

y-cut $x = 0$ in the original: 5

$a > 0$: cup, minimum. $a < 0$: cap, maximum.



Sketch questions

Mark three things and nothing else: the turning point, the y-cut, and the right shape (cup or cap).

CALCULATOR CHECK On a ClassWiz (fx-991EX / CW): MENU → Equation → Polynomial → 2, enter a, b, c and keep pressing = past the roots — it shows the *x-Value Minimum / Maximum* and the *y-Value*. Get the answer first, then write the working.

IDEA 03

Practise: Read the Turning Point

My questions first, then selected practice, then real past papers.

Q19**DR ESLAM**

Read it straight off

CH 2 · TOPIC 5 3 marks

The curve C has equation $y = 3(x - 1)^2 + 2$.

Write down (a) the turning point, (b) whether it is a maximum or a minimum, (c) the line of symmetry, (d) where C cuts the y -axis.

Q20**DR ESLAM**

Complete, then read

CH 2 · TOPIC 5 3 marks

Write $x^2 - 10x + 4$ in the form $(x - p)^2 - q$.

Hence write down the turning point of $y = x^2 - 10x + 4$.

Turning point _____

SELECTED PRACTICE

Q21**PRACTICE**

Practice · Medium

CH 2 · TOPIC 5 1 mark

4 The equation of a curve is $y = (x + 3)^2 + 5$

Circle the coordinates of the turning point.

(5, 3)	(5, -3)	(3, 5)	(-3, 5)
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(1 mark)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5**3 marks**

2 Given that $x^2 - 6x + 1 = (x - a)^2 - b$ for all values of x .

(i) Find the value of a and the value of b .

[2]

(ii) Hence write down the coordinates of the turning point of the graph of $y = x^2 - 6x + 1$.

[1]

(3 marks)

Q23**PRACTICE**

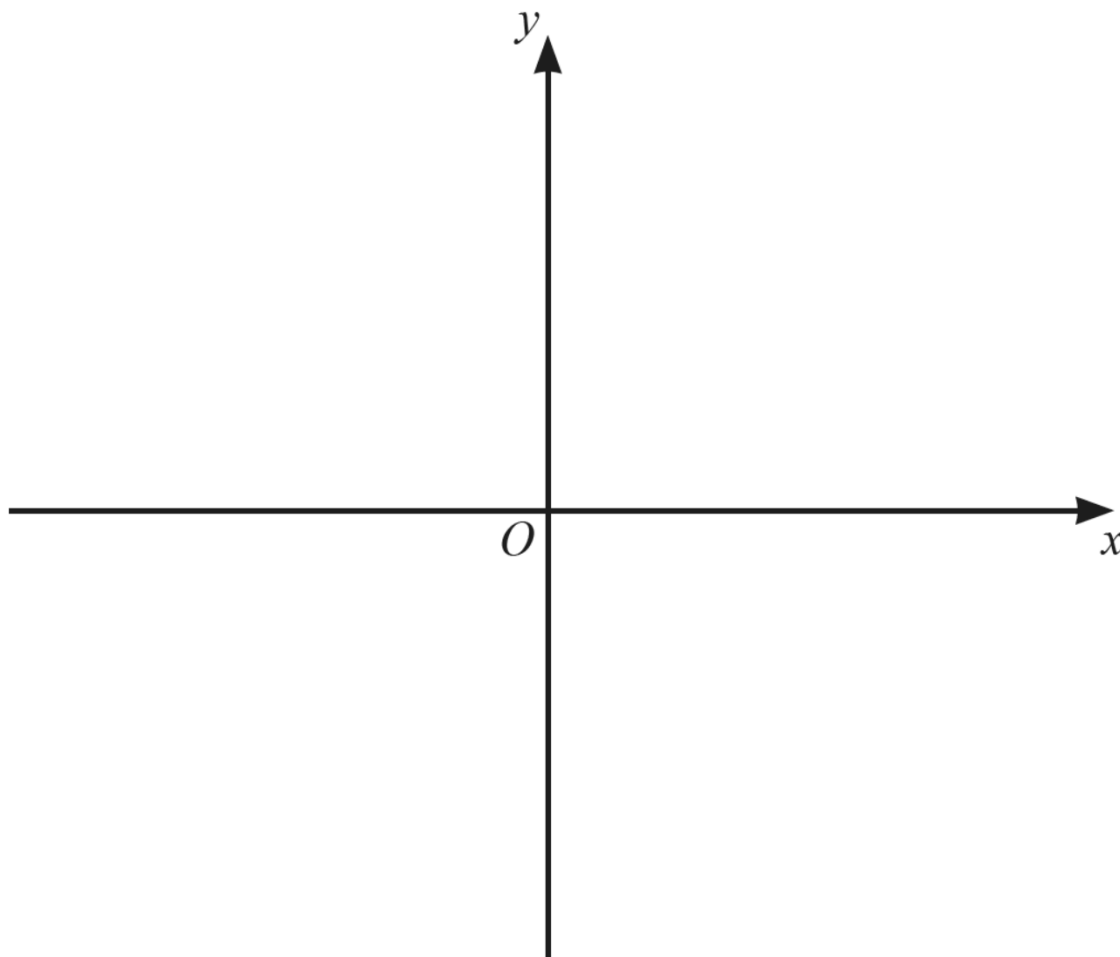
Practice · Hard

CH 2 · TOPIC 5 5 marks

12 (a) Write $x^2 + 10x + 14$ in the form $(x + a)^2 + b$.

(2 marks)

(b) Sketch the graph of $y = x^2 + 10x + 14$, indicating the coordinates of the turning point.

**(3 marks)**

Q24**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 5 4 marks

8 Find the turning point of $y = x^2 + 4x - 3$ by completing the square.

(..... ,)

(4 marks)

PAST PAPERS

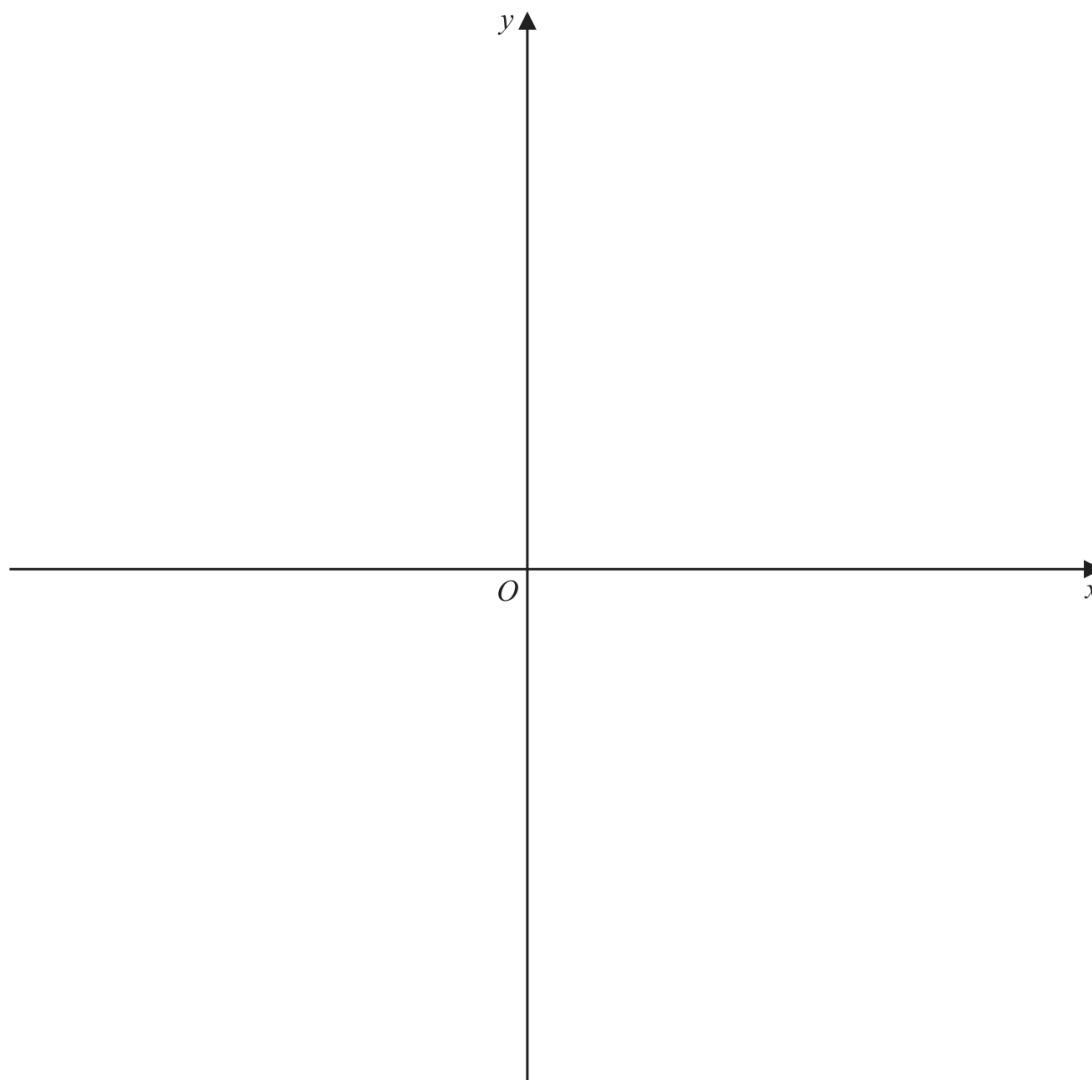
Q25 PAST PAPER Edexcel 4MA1 · June 2025 · Paper 2H · Q25

CH 2 · TOPIC 5 6 marks

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(Total for Question 25 is 6 marks)

04 The “Hence” Questions

Part (a) is always the completed square. Part (b) uses it — there are only three things it can ask.

WHEN YOU SEE → YOU DO

“hence solve”

→ Move → Root ($\pm$) → Solve

“surd form” / “exact”

→ keep the $\sqrt{}$, do not round

“ $y = f(x + 4)$ ”

→ find the turning point of f , then move it

“series of transformations ... maps S onto T”

→ translate · stretch · translate

A

Hence solve = Move → Root → Solve

$$x^2 + 4x - 16 = 0 \Rightarrow (x + 2)^2 - 20 = 0$$

$$\text{Move: } (x + 2)^2 = 20 \quad \text{Root: } x + 2 = \pm\sqrt{20} \quad \text{Solve: } x = -2 \pm 2\sqrt{5}$$

Never forget the $\pm$ — two answers.

B

The curve is moved: $y = f(x + 4)$

Find the turning point of f first, then move it.

$f(x) = 2(x - 3)^2 - 15$ has minimum $(3, -15)$. $f(x + 4)$ moves it **4 left** → $(-1, -15)$.

Inside the bracket moves x the opposite way; outside moves y the same way.

C

Transformations from $y = x^2$

$a(x + p)^2 + q$: translate $\begin{pmatrix} -p \\ 0 \end{pmatrix}$ → stretch parallel to the y -axis, s.f. a →

translate $\begin{pmatrix} 0 \\ q \end{pmatrix}$. Order matters.

$2(x - 3)^2 - 5$: translate $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, stretch s.f. 2, then translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$.

IDEA 04

Practise: The “Hence” Questions

My questions first, then selected practice, then real past papers.

Q26**DR ESLAM**

Move → Root → Solve

CH 2 · TOPIC 5 4 marks

(a) Write $x^2 - 4x - 5$ in the form $(x - p)^2 - q$.

(b) Hence solve $x^2 - 4x - 5 = 0$.

$x =$ _____

Q27**DR ESLAM**

The curve is moved

CH 2 · TOPIC 5 2 marks

$$f(x) = (x - 2)^2 - 9.$$

Write down the coordinates of the turning point of the curve $y = f(x + 3)$.

Turning point _____

SELECTED PRACTICE

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5 7 marks

8 (i) Write $x^2 + 4x - 16$ in the form $(x + a)^2 - b$.

[3]

(ii) Solve the equation $x^2 + 4x - 16 = 0$.

Give your answers in surd form as simply as possible.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

(7 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5 4 marks

10 (i) Write $x^2 + 8x - 9$ in the form $(x + k)^2 + h$.

[2]

(ii) Use your answer to **part (i)** to solve the equation $x^2 + 8x - 9 = 0$.

 $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]

(4 marks)

Q30**PRACTICE**

Practice · Hard

CH 2 · TOPIC 5 4 marks

11 (a) Write $x^2 - 18x - 27$ in the form $(x + k)^2 + h$.

(2 marks)

(b) Use your answer to **part (a)** to solve the equation $x^2 - 18x - 27 = 0$.

 $x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 2H · Q22

CH 2 · TOPIC 5 4 marks

- 22 The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants,
describe fully a series of transformations that map the curve **S** onto the curve **T**.

.....

.....

.....

(Total for Question 22 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q23

CH 2 · TOPIC 5 5 marks

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

.....

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.....,)

(Total for Question 23 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q20

CH 2 · TOPIC 5 5 marks

- 20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

- (b) Find the coordinates of P

(.....,)

(Total for Question 20 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

$$\text{List } a, b, c, \frac{b}{2a} \rightarrow a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$$

01 Two-step method

02 Negative a

Reorder to $ax^2 + bx + c$; carry the minus. Write the number first.

03 Turning point

$a(x + p)^2 + q \rightarrow (-p, q)$. $a > 0$ min, $a < 0$ max.
Symmetry $x = -p$.

04 “Hence”

Solve: Move $\rightarrow$ Root ($\pm$) $\rightarrow$ Solve. $f(x+k)$: move k left. Transform: translate, stretch, translate.

BOTTOM LINE List $a, b, c, \frac{b}{2a} \rightarrow$ structure $\rightarrow$ calculator on the right-hand part. Turning point: flip inside, keep outside. “Hence” = solve, move, or transform.

Answers

Completing the Square • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE TWO-STEP METHOD

- | | | |
|-----------|---|---|
| Q1 | Dr Eslam • Step 1 first • $a = 1$ | $(x + 3)^2 + 2$ |
| Q2 | Dr Eslam • $a \neq 1$ • the trap | $2(x + 3)^2 - 11$ |
| Q3 | Practice • Medium | $(x + 3)^2 - 16$ |
| Q4 | Practice • Medium | $(x - 3)^2 + 2$ |
| Q5 | Practice • Medium | $a = 2, b = -13$ |
| Q6 | Practice • Very Hard | $a = 2, b = \frac{3}{2}, c = \frac{1}{2}$ |
| Q7 | Edexcel 4MA1 • January 2022 • Paper 1HR • Q19 | (a) $m = -\frac{4}{3}$ (b) $3(x + 2)^2 + 7$ |
| Q8 | Edexcel 4MA1 • June 2023 • Paper 2H • Q19 | $3(x - 1)^2 + 2$ |
| Q9 | Edexcel 4MA1 • November 2025 • Paper 1H • Q21 | $5(x - 2)^2 + 3$ |

IDEA 02 • NEGATIVE a

- Q10** Dr Eslam • Negative $a = -1$ $19 - (x + 3)^2$, so $p = 19$, $q = 3$
- Q11** Dr Eslam • Negative $a \neq -1$ $13 - 2(x - 2)^2$, so $a = 13$, $b = 2$, $c = 2$
- Q12** Practice • Very Hard $14 - (x - 3)^2$: $p = 14$, $q = 3$
- Q13** Practice • Very Hard $23 - 2(x - 3)^2$: $a = 23$, $b = -2$, $c = -3$
- Q14** Practice • Very Hard $11 - (x + 2)^2$: $p = 11$, $q = 2$
- Q15** Edexcel 4MA1 • January 2020 • Paper 2HR • Q23 $25 - 2(x + 3)^2$
- Q16** Edexcel 4MA1 • January 2022 • Paper 1H • Q20 (a) $19 - 3(x - 2)^2$ (b) $A(2, 19)$
- Q17** Edexcel 4MA1 • June 2022 • Paper 1H • Q24 $a = q + \frac{36}{q}$, $b = q$, $c = \frac{6}{q}$
- Q18** Edexcel 4MA1 • May 2024 • Paper 1H • Q25 $29 - 3(x - 2)^2$

IDEA 03 • READ THE TURNING POINT

- Q19** Dr Eslam • Read it straight off (a) $(1, 2)$ (b) minimum (c) $x = 1$ (d) $(0, 5)$
- Q20** Dr Eslam • Complete, then read $(x - 5)^2 - 21$; turning point $(5, -21)$
- Q21** Practice • Medium $(-3, 5)$
- Q22** Practice • Hard (i) $a = 3$, $b = 8$ (ii) $(3, -8)$
- Q23** Practice • Hard (a) $(x + 5)^2 - 11$ (b) U-shape, min $(-5, -11)$, y-cut $(0, 14)$
- Q24** Practice • Very Hard $(-2, -7)$
- Q25** Edexcel 4MA1 • June 2025 • Paper 2H • Q25 (a) $52 - 6(x - 2)^2$ (b) cap shape, max $(2, 52)$, y-cut $(0, 28)$

IDEA 04 • THE “HENCE” QUESTIONS

- Q26** Dr Eslam · Move → Root → Solve
(a) $(x - 2)^2 - 9$ (b) $x = 5$ or $x = -1$
- Q27** Dr Eslam · The curve is moved
 $(-1, -9)$
- Q28** Practice · Hard
(i) $(x + 2)^2 - 20$ (ii) $x = -2 \pm 2\sqrt{5}$
- Q29** Practice · Hard
(i) $(x + 4)^2 - 25$ (ii) $x = 1$ or $x = -9$
- Q30** Practice · Hard
(a) $(x - 9)^2 - 108$ (b) $x = 9 \pm 6\sqrt{3}$
- Q31** Edexcel 4MA1 · May 2021 · Paper 2H · Q22
 $g(x) = 2(x - 3)^2 - 5$: translate $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, stretch || y-axis s.f. 2, translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$
- Q32** Edexcel 4MA1 · June 2022 · Paper 1HR · Q23
(a) $2(x - 3)^2 - 15$ (b) $M(-1, -15)$
- Q33** Edexcel 4MA1 · May 2024 · Paper 1HR · Q20
(a) $2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$ (b) $P\left(\frac{23}{4}, -\frac{49}{8}\right)$